

## 09.01 Right Triangle Trigonometry

Practice problems

Show your work and state restrictions or excluded values when needed.

| Recognize                                                    | Apply these rules                                                                                              |
|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Trigonometric Functions And Models                           | Identify the angle measure, quadrant, or graph feature before selecting a formula.                             |
| The expression contains the chapter's key notation or model. | Write the known quantities, choose the matching formula, calculate carefully, and check the result in context. |

### A. Practice: Right Triangle Trigonometry

1.

$$\sin 30^\circ = \frac{1}{2}$$

2.

$$135^\circ \cdot \frac{\pi}{180}$$

3.

$$y = 3\sin\left(\frac{\pi x}{2}\right) + 4$$

4.

$$\sin 30^\circ = \frac{1}{2}$$

5.

$$135^{\circ} \cdot \frac{\pi}{180}$$

6.

$$y = 3\sin\left(\frac{\pi x}{2}\right) + 4$$

7.

$$\sin 30^{\circ} = \frac{1}{2}$$

8.

$$135^{\circ} \cdot \frac{\pi}{180}$$

9.

$$y = 3\sin\left(\frac{\pi x}{2}\right) + 4$$

10.

$$\sin 30^{\circ} = \frac{1}{2}$$

11.

$$135^{\circ} \cdot \frac{\pi}{180}$$

12.

$$y = 3\sin\left(\frac{\pi x}{2}\right) + 4$$

13.

$$\sin 30^{\circ} = \frac{1}{2}$$

14.

$$135^{\circ} \cdot \frac{\pi}{180}$$

15.

$$y = 3\sin\left(\frac{\pi x}{2}\right) + 4$$

16.

$$\sin 30^{\circ} = \frac{1}{2}$$

17.

$$135^{\circ} \cdot \frac{\pi}{180}$$

18.

$$y = 3\sin\left(\frac{\pi x}{2}\right) + 4$$

19.

$$\sin 30^{\circ} = \frac{1}{2}$$

20.

$$135^{\circ} \cdot \frac{\pi}{180}$$

### B. Apply the key formulas

1.

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

2.

$$B = \frac{2\pi}{T}$$

3.

$$y = A\sin(Bc) + D$$

4.

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

5.

$$B = \frac{2\pi}{T}$$

6.

$$y = A\sin(Ba) + D$$

7.

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

8.

$$B = \frac{2\pi}{T}$$

9.

$$y = A\sin(Bd) + D$$

10.

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

11.

$$B = \frac{2\pi}{T}$$

12.

$$y = A\sin(Bb) + D$$

13.

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

14.

$$B = \frac{2\pi}{T}$$

15.

$$y = A\sin(Be) + D$$

16.

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

17.

$$B = \frac{2\pi}{T}$$

18.

$$y = A\sin(Bc) + D$$

19.

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

20.

$$B = \frac{2\pi}{T}$$